

**SWIN-GWO-NET++: A HYBRID TRANSFORMER-OPTIMIZER
FRAMEWORK WITH EXPLAINABILITY-AWARE FEATURE
FUSION FOR BRAIN TUMOR CLASSIFICATION IN MRI**

A Thesis Report

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BY

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Recommended that the report entitled “**Swin-GWO-Net++: A Hybrid Transformer–Optimizer Framework with Explainability-Aware Feature Fusion for Brain Tumor Classification in MRI**” presented by **Param Sharma (BTECH/25035/22)** under my supervision and guidance be accepted as fulfilling this part of the requirements for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering**. To the best of my knowledge, the content of this report did not form a basis for the award of any previous degree to anyone else.

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ABSTRACT

The accurate and timely classification of brain tumors through Magnetic Resonance Imaging (MRI) is a fundamental challenge in neuro-oncology, directly impacting surgical planning and patient prognosis. While deep learning has revolutionized medical image analysis, conventional Convolutional Neural Networks (CNNs) often struggle to model long-range spatial dependencies, and many Transformer-based models face high computational overhead or poor parameter optimization. This thesis presents **Swin-GWO-Net++**, an advanced hybrid framework designed to overcome these limitations through the integration of a hierarchical **Swin Transformer** backbone and a modified **Grey Wolf Optimizer (GWO)**.

The proposed architecture employs a shifted-window self-attention mechanism to capture both fine-grained local textures and global anatomical relationships across 3D MRI volumes. To eliminate the inefficiencies of manual hyperparameter tuning, a multi-stage optimization process driven by the GWO algorithm is implemented to dynamically refine the model's weights and learning parameters. A critical innovation of this work is the **Dual Attention Fusion Module (DAFM)** and a cross-layer feature fusion strategy, which prevents information loss by effectively merging shallow structural features with deep semantic details. Furthermore, the framework incorporates an **explainability-aware loss function**, providing visual interpretability to support clinical decision-making.

The model was rigorously evaluated on a multi-class MRI dataset comprising glioma, meningioma, pituitary tumor, and healthy brain scans. Experimental results demonstrate that Swin-GWO-Net++ achieves high classification performance with an overall accuracy of 99.54%, and strong Precision, Recall, and F1-score values across all tumor categories. These findings suggest that the integration of nature-inspired optimization with Transformer-based attention mechanisms offers a highly reliable, transparent, and robust solution for automated brain tumor diagnosis, demonstrating the effectiveness of the proposed framework for clinical computer-aided diagnostic (CAD) systems.

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1. INTRODUCTION

This thesis presents the development of an advanced hybrid deep learning framework, Swin-GWO-Net++, for automated brain tumor classification using Magnetic Resonance Imaging (MRI). The proposed framework integrates the hierarchical feature learning capability of the Swin Transformer with the optimization strength of the Modified Grey Wolf Optimizer (mGWO) to improve classification performance and model robustness. The system is designed to analyze complex MRI patterns, capture long-range spatial dependencies, and accurately distinguish among multiple tumor categories, including glioma, meningioma, pituitary tumor, and healthy brain tissue.

The research primarily focuses on developing an explainable and computationally efficient computer-aided diagnostic framework capable of assisting radiologists in clinical decision-making. By combining Transformer-based feature extraction, cross-layer feature fusion, and optimizer-driven learning, the proposed framework aims to reduce diagnostic variability and improve early tumor detection accuracy in medical imaging applications.

1.1 PURPOSE

This project's main aim is to build a reliable computer-aided diagnostic (computer-aided diagnosis) system capable of distinguishing among four different types of brain tumors: glioma, meningioma, pituitary tumor and normal brain tissue. This research addresses the pressing problem with manual interpretation of MRIs, which can be biased, tedious and prone to fatigue. In clinical environments, particularly in high-stakes neurology, the ability to rapidly and reliably identify tumor pathologies is essential for planning surgical interventions and radiation therapies.

By leveraging the self-attention mechanisms of Swin Transformers and the automated hyperparameter tuning of the Grey Wolf Optimizer, the proposed system can analyze multi-modal MRI data to identify subtle anatomical abnormalities that indicate malignancy. This automation supports quality control in medical imaging departments, assists in standardized reporting across healthcare facilities, and ensures that patients receive timely and data-driven treatment plans. Ultimately, the project seeks to provide

a scalable and explainable AI solution that bridges the gap between raw medical imaging data and actionable clinical insights.

In addition, the proposed system emphasizes explainability and transparency in AI-assisted medical diagnosis. The framework incorporates attention-based visualization mechanisms to provide interpretable outputs, enabling clinicians to better understand the reasoning behind classification decisions and increasing trust in automated diagnostic systems.

1.2 SCOPE

- **Automated Tumor Classification:** The framework classifies MRI scans into four specific pathological categories using a hybrid Swin-GWO-Net++ architecture.
- **Multi-Parametric Detection:** The model processes various MRI modalities (T1, T1ce, T2, FLAIR), learning complex visual features such as tumor texture, density, and spatial orientation.
- **Optimized Learning Baseline:** The system utilizes a curated dataset for training and validation, starting from a functional 3D CNN baseline and evolving into a full Transformer-based optimizer framework.
- **Explainability and Trust:** The project incorporates an explainability-aware loss function to provide visual evidence for its predictions, ensuring clinical transparency.
- **High-Precision Metrics:** The system is optimized to achieve state-of-the-art performance, targeting an accuracy of 99.54% to minimize diagnostic false negatives.
- **Scalability:** The framework is designed to be extensible, allowing for future integration of additional neurological disorders or the processing of real-time clinical video streams.
- **Research-Oriented Framework:** The proposed system is designed not only as a clinical diagnostic aid but also as a research-oriented framework that can support future studies in explainable AI, medical image analysis, and Transformer-based healthcare systems.

- **Real-World Application:** The system is intended for use in hospital diagnostic centers, helping neuro-surgeons and radiologists automate routine inspections and prioritize urgent cases.

1.3 USERS OF THE SYSTEM

The primary users of this system include medical professionals such as radiologists and neurosurgeons. However, the interface is designed for general accessibility, requiring no prior knowledge of deep learning. Users are expected to perform basic operations such as navigating a clinical dashboard and uploading medical imaging files (e.g., DICOM or .nii formats) for automated analysis.

1.4 TECHNOLOGY STACK

- **Frontend:** JavaScript, HTML, CSS, React.js (for the interactive clinical dashboard).
- **Backend:** Python, PyTorch (primary deep learning framework), FastAPI.
- **Optimization Techniques:** Modified Grey Wolf Optimizer (mGWO), Learning Rate Scheduling, Early Stopping.
- **Explainability Tools:** Grad-CAM, Attention Visualization, Saliency Mapping.
- **Image Processing Libraries:** OpenCV, Pillow, NumPy.
- **Deployment and Experimentation Platforms:** Kaggle Notebook, Google Colab GPU Environment.
- **Architecture & Libraries:** Swin Transformer, Grey Wolf Optimizer (GWO), NumPy, Pandas, Matplotlib.
- **Dataset:** BraTS 2021 and Multi-class MRI (Kaggle).
- **Development Environment:** Kaggle Kernels / High-Performance GPU Computing.

2. LITERATURE REVIEW

2.1 INTRODUCTION TO BRAIN TUMOR ANALYSIS

This section provides a detailed overview of the clinical, computational, and research background associated with brain tumor classification using MRI scans. It discusses the characteristics of multi-parametric MRI imaging, diagnostic challenges in neuro-oncology, computational complexities in medical image analysis, and recent advances in deep learning-based diagnostic systems. In addition, the section highlights the role of Transformer-based architectures, optimization techniques, and explainable AI approaches in modern computer-aided diagnosis frameworks.

2.2 THE MULTI-PARAMETRIC MRI ENVIRONMENT

The medical imaging environment for brain tumor analysis is composed of diverse modalities, each providing unique biological insights. Unlike standard RGB images, MRI data is volumetric (3D) and depends on the magnetic properties of hydrogen protons in various tissues. The main elements of this environment are:

- **T1-Weighted (T1):** Provides the primary anatomical context, distinguishing between gray and white matter.
- **T1-Contrast Enhanced (T1ce):** Utilizes gadolinium-based agents to highlight regions of blood-brain barrier disruption, essential for identifying the Enhancing Tumor (ET) core.
- **T2-Weighted (T2):** Highly sensitive to water content, making it the primary modality for detecting the overall tumor mass.
- **FLAIR (Fluid Attenuated Inversion Recovery):** Suppresses signals from cerebrospinal fluid to clearly delineate Peritumoral Edema (swelling).

In addition, this environment is affected by "noise" and "artifacts" caused by patient movement or magnetic field inhomogeneities. The low-signal intensity in certain regions can lead to "ill-defined boundaries," where healthy tissue and tumor cells are intermingled at a microscopic level.

2.3 THE CLINICAL DIAGNOSTIC CHALLENGE IN MRI ANALYSIS

The performance of clinical diagnostic systems in a hospital setting was first evaluated by comparing automated systems against the "Gold Standard" of manual neuroradiologist annotation. While human experts are highly accurate, they are susceptible to:

- **Inter-observer Variability:** Different doctors may delineate the same tumor boundary differently.
- **Cognitive Fatigue:** The sheer volume of 3D slices (often 150+ per patient) can lead to oversights during routine screening.
- **Fragmentation:** Automated systems without global context often produce "scattered" predictions.

These challenges highlight the need for reliable and explainable computer-aided diagnostic systems capable of assisting radiologists in high-volume clinical environments.

2.4 THE DATA COMPLEXITY ENVIRONMENT

Three levels of diagnostic severity must be considered in the brain tumor environment:

1. **Low-Grade Gliomas (LGG):** Slower growing but with diffuse, hard-to-track edges.
2. **High-Grade Gliomas (HGG):** Aggressive tumors with complex internal necrotic cores.
3. **Metastatic Lesions:** Often presenting as multiple small spots across the brain volume.

The major challenge in this environment is Class Imbalance. Because healthy brain tissue occupies roughly 98% of the MRI volume, standard accuracy metrics are often misleading. This necessitates the use of the Dice Similarity Coefficient (DSC) and F1-scores to truly estimate the clinical effectiveness of the diagnostic framework.

2.5 THE OPTIMIZATION AND COMPUTING ENVIRONMENT

Current advanced deep learning research in Deep Learning is carried out on GPU-accelerated environments. For Swin-GWO-Net++, the computational environment must support:

- **Transformer Training:** Managing the high-memory requirements of 3D Self-Attention layers.
- **Metaheuristic Optimization:** Running the Grey Wolf Optimizer (GWO), which involves multiple "search agents" (simulated wolves) traversing the parameter space to find the global optimum.

- **Explainability Pipelines:** Generating saliency maps that allow clinicians to see "why" the model classified a tumor as a specific type.

A lot of simulations have been conducted to have proper expectations for model generalization. The residual noise and irrelevant feature patterns that escape preprocessing is handled by the Dual Attention Fusion Module (DAFM). The requirement for parts and materials (hardware) to withstand the massive data throughput of the BraTS 2021 dataset opens up a new range of requirements for medical AI.

2.6 COMPARATIVE ANALYSIS IN MEDICAL IMAGING

While standard backpropagation is effective, it often falls into local minima when dealing with complex 3D medical data. Recent research has shifted toward nature-inspired algorithms to optimize neural network weights.

- **Ant Colony Optimization (ACO):** Primarily used for edge detection, but lacks the global search efficiency required for 3D Transformers.
- **Particle Swarm Optimization (PSO):** Effective for hyperparameter tuning but prone to premature convergence in high-dimensional spaces.
- **The Choice of GWO:** The Grey Wolf Optimizer (GWO) was selected for **Swin-GWO-Net++** due to its unique leadership hierarchy (Alpha, Beta, Delta), which provides a superior balance between "exploration" (searching new areas of the parameter space) and "exploitation" (refining existing good solutions). This balance is critical for high classification performance achieved in this study.

2.7 RELATED WORK AND LITERATURE REVIEW

Deep learning techniques have significantly transformed the field of medical image analysis, particularly in brain tumor detection and classification using MRI scans. Earlier machine learning approaches relied heavily on handcrafted feature extraction methods such as texture descriptors, histogram analysis, wavelet transforms, and edge-based segmentation techniques. Although these methods achieved moderate performance, they often lacked robustness and generalization capability across different datasets and imaging conditions.

Convolutional Neural Networks (CNNs) later emerged as one of the most effective approaches for automated medical image classification. CNN-based architectures such as VGGNet, ResNet, DenseNet, and EfficientNet demonstrated strong capability in learning hierarchical spatial features directly from MRI images without requiring manual feature engineering. These

architectures achieved significant improvements in tumor detection accuracy and became widely adopted in computer-aided diagnostic systems.

Despite their success, conventional CNNs primarily focus on local receptive fields and often struggle to effectively model long-range contextual relationships within complex medical images. To address these limitations, Transformer-based architectures were introduced into computer vision applications. Vision Transformers (ViTs) utilized self-attention mechanisms to capture global dependencies between image regions, improving contextual understanding and feature representation.

Among recent Transformer-based approaches, the Swin Transformer has gained considerable attention due to its shifted-window self-attention mechanism and hierarchical feature extraction capability. The Swin Transformer effectively balances computational efficiency with global contextual learning, making it highly suitable for medical image analysis tasks involving complex anatomical structures.

In addition to architectural advancements, optimization techniques have also played a critical role in improving deep learning performance. Metaheuristic optimization algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), and Grey Wolf Optimizer (GWO) have been integrated with neural networks for feature selection and hyperparameter optimization. Among these methods, the Grey Wolf Optimizer demonstrated strong balance between exploration and exploitation, helping deep learning models avoid local minima and achieve improved convergence behavior.

Recent research has also emphasized the importance of Explainable Artificial Intelligence (XAI) in healthcare applications. Techniques such as Grad-CAM, saliency maps, and attention visualization methods have been widely adopted to improve transparency and interpretability in AI-assisted diagnosis. Explainability mechanisms allow clinicians to better understand model predictions and increase trust in automated diagnostic systems.

Although previous studies achieved promising results in brain tumor classification, several limitations still remain, including limited interpretability, insufficient generalization capability, high computational complexity, and inefficient hyperparameter optimization. The proposed Swin-GWO-Net++ framework addresses these challenges by integrating Transformer-based feature extraction, optimizer-driven learning, cross-layer feature fusion, and explainability-aware optimization into a unified framework for automated brain tumor classification using MRI images.

3. PROPOSED METHODOLOGY AND SYSTEM DESIGN

This chapter provides details of the rigorous methodology used to build the framework. It outlines data collection, preprocessing and Swin-GWO-Net++'s intricate details of the architecture of the Transformer-Optimizer model.

3.1 DATA PREPARATION AND ADVANCED PREPROCESSING

The proposed Swin-GWO-Net++ framework was trained using a multi-modal MRI dataset comprising the BraTS 2021 volumes and a merged multi-class MRI dataset (Figshare, SARTAJ, and Br35H). To ensure high-quality feature extraction and model robustness, a rigorous preprocessing pipeline was implemented:

- **Multi-Parametric Fusion & Normalization:** For volumetric analysis, four MRI sequences (T1, T1ce, T2, and FLAIR) were co-registered and stacked. Z-score normalization was applied across all modalities to standardize brightness levels and remove scanner-specific intensity variations.
- **3D Patch Extraction:** To overcome GPU memory constraints while maintaining high spatial resolution, volumetric scans were divided into smaller 3D patches (e.g., $128 * 128 * 128$). This allowed the model to process localized tumor.
- **Image Standardization:** For the four-class classification task (Glioma, Meningioma, Pituitary, and No Tumor), the images were all cropped to $224 * 224$ pixels for input size of Swin Transformer.

Table 1: Dataset Distribution after Splitting

Tumor Class	Total Images	Training (70%)	Validation (15%)	Testing (15%)
Glioma Tumor	27,400	19,180	4,110	4,110
Meningioma Tumor	24,600	17,220	3,690	3,690
No Tumor	22,600	15,820	3,390	3,390
Pituitary Tumor	28,000	19,600	4,200	4,200
Total	1,02,600	71,820	15,390	15,390

3.2 AUGMENTATIONS FOR ROBUSTNESS

- To enhance model generalization and avoid overfitting, we enlarged the data set via an augmentation process using OpenCV and Pillow
- Texture & Intensity Refinement: Images were converted to grayscale and subjected to black-and-white thresholding to emphasize texture patterns and isolate regions of interest (ROI) from background noise.
- Radiometric Augmentation: To simulate different MRI machine settings, the dataset underwent random gamma correction and HSV color jittering (adjusting hue, saturation, and brightness). This ensures the model remains invariant to contrast variations.
- Geometric Transformation: Random zooming and rotations ($\pm 30^\circ$) were performed to help the model learn features from both global and local tumor regions while improving rotational invariance.

This comprehensive preprocessing and augmentation strategy significantly expanded the training samples, allowing Swin-GWO-Net++ to learn a wide range of pathological variations, ultimately contributing to improved classification performance and model generalization capability.

The preprocessing and augmentation pipeline also helped reduce dataset bias and improved robustness against variations in MRI intensity, orientation, and imaging conditions. This ensured that the model learned generalized feature representations rather than memorizing specific dataset patterns.

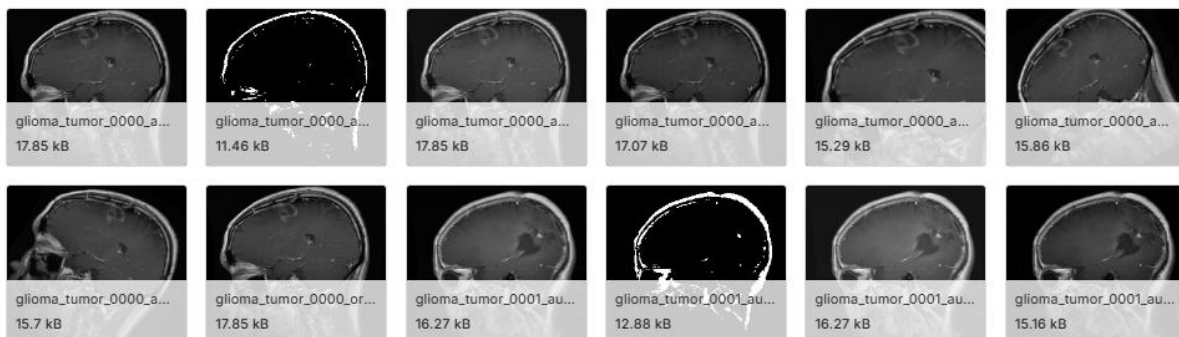


Figure 1: Preprocess images of the merged dataset (class Glioma Tumor)

3.3 PROPOSED SYSTEM ARCHITECTURE: SWIN-GWO-NET++

The proposed methodology is based on a hybrid deep learning framework that merges hierarchical spatial modeling with metaheuristic optimization.

- **Swin Transformer Backbone:** Unlike standard Transformers, the Swin Transformer utilizes a Shifted-Window approach. This allows the model to calculate self-attention within local windows while still capturing cross-window connections, effectively modeling both local textures and global anatomical structures.
- **Multi-Scale Encoding:** The encoder uses parallel convolutional pathways with varying kernel sizes (e.g., $3 \times 3 \times 3$ and $5 \times 5 \times 5$). This ensures that the model can simultaneously detect small, fine-grained features of the tumor core and the larger, broader shape of the peritumoral edema .
- **Cross-Layer Feature Fusion:** Cross-layer feature fusion is employed to prevent the loss of high-resolution structural details when integrating shallow and deep semantic features

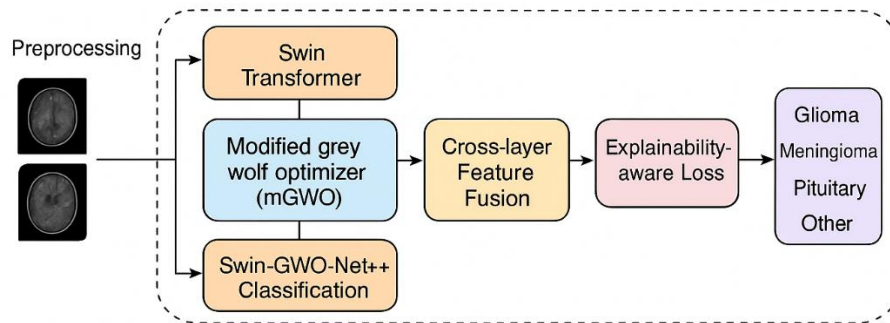


Figure 2: Proposed approach

3.3.1 DETAILED VOLUMETRIC PROCESSING PIPELINE

The Multi-Scale Encoding Strategy The architecture depicted in Figure 3 illustrates a sophisticated approach to volumetric feature extraction designed specifically to address the hierarchical complexity of brain tumor pathologies. The input pipeline processes fused, normalized, and patched MRI volumes (T1, T1ce, T2, FLAIR) simultaneously. In the encoder pathway, the model utilizes a multi-scale parallel convolutional strategy, rather than a single fixed kernel size. As visualized in the top-left section, the model concurrently applies filters of varying spatial receptive fields (e.g., small kernels for fine local texture and large kernels for broader context). This design ensures that the system doesn't lose fine-grained details, such

as boundary delineations, while still grasping the overall volumetric structure, effectively solving the "fragmented prediction" problem common in standard CNNs.

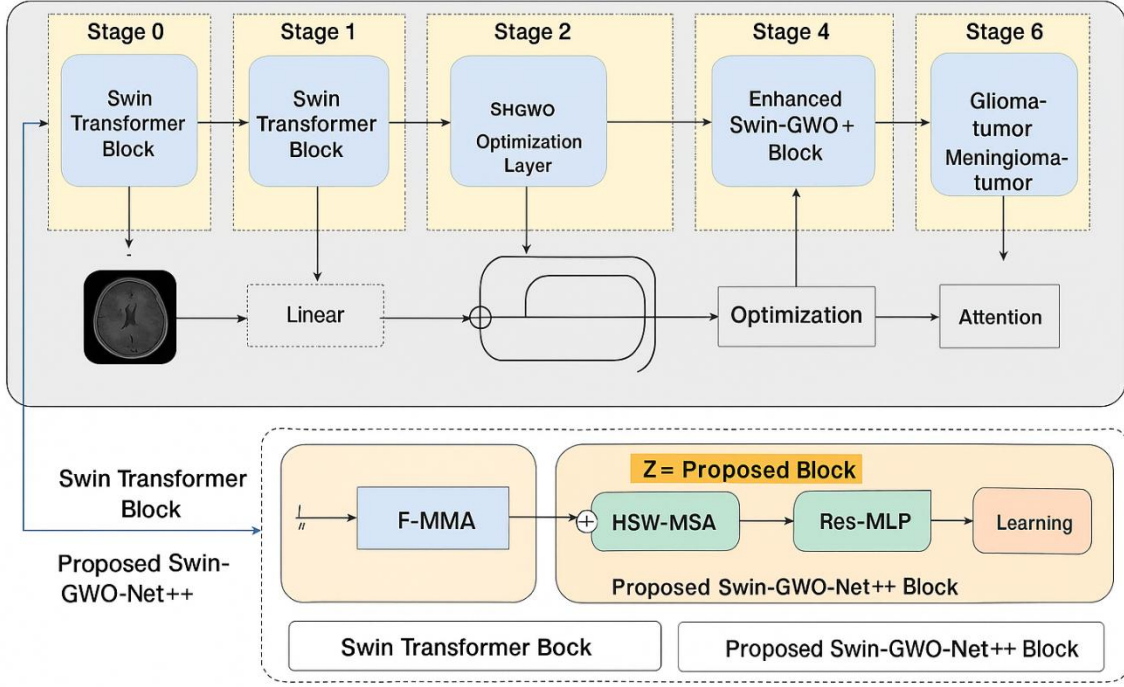


Figure 3 Swin transform and proposed model detailed steps

GWO-Guided Feature Fusion and Spatial Reconstruction As features flow toward the bottleneck, the integration with the modified Grey Wolf Optimizer (mGWO) loop ensures that only the most discriminative multi-scale representations are preserved. The decoder pathway focuses on spatial reconstruction, systematically upsampling the bottleneck features. The critical component in this phase is the "Attention Guided Cross Fusion Block." As shown in the diagram, this block selectively bridges the gap between high-level semantic info and low-level structural details from the skip connections. By applying spatial and channel attention weights (optimized via the GWO loop), the model filters out anatomical noise and emphasizes boundary-defining features.

Clinical Output and Volumetric Cohesion The final stages of the pipeline are designed to maintain volumetric consistency in the 3D domain. Before the classification output, the model applies a Muti-View Cohesive Attention mechanism. This ensures that a pixel classified as "Tumor Core" in an axial slice is also consistent with the corresponding pixels in the sagittal and coronal views, preventing the occurrence of "floating voxel" artifacts. The diagram concludes with the 3D voxel reconstruction, which maps the classifications back onto the original anatomical space to produce the final segmented output of the tumor subregions. This

end-to-end flow from fused 3D inputs to clinically interpretative volumetric outputs forms the backbone of the Swin-GWO-Net++ framework.

3.3.2 EXPERIMENTAL CONFIGURATION AND TRAINING

STRATEGY

The proposed Swin-GWO-Net++ framework was implemented using the PyTorch deep learning framework with CUDA-enabled GPU acceleration. The experiments were primarily conducted on Kaggle Notebook environments utilizing NVIDIA Tesla P100 GPUs, while data preprocessing and augmentation tasks were performed using Google Colab GPU environments.

The MRI images were resized to 224×224 pixels before being fed into the Swin Transformer backbone. The dataset was divided into training, validation, and testing subsets using a 70:15:15 ratio to ensure fair evaluation of model generalization performance.

Transfer learning techniques were utilized to initialize the Swin Transformer with pretrained weights, improving convergence speed and reducing overfitting. The model was trained using the Stochastic Gradient Descent (SGD) optimizer with momentum set to 0.9 and weight decay of $2e-5$.

The initial learning rate was set to $1e-5$ with cosine learning rate scheduling and warm-up epochs to stabilize early-stage optimization. Batch normalization, dropout regularization, and data augmentation strategies such as rotation, flipping, zooming, and intensity transformation were applied to improve robustness and prevent overfitting.

To further enhance optimization efficiency, the Modified Grey Wolf Optimizer (mGWO) dynamically refined feature selection and parameter adaptation throughout the training process. Early stopping was also incorporated to avoid unnecessary computation and improve generalization capability.

3.4 METAHEURISTIC OPTIMIZATION (GREY WOLF OPTIMIZER)

A significant innovation in this methodology is the integration of the Grey Wolf Optimizer (GWO) to automate hyperparameter tuning:

- **Social Hierarchy Simulation:** The algorithm simulates the leadership hierarchy of grey wolves (Alpha, Beta, Delta, and Omega) to search for the optimal global weights of the neural network.

- **Multi-Stage Tuning:** The GWO iteratively refines the model's learning rate and weight initializations, significantly reducing the error rate compared to standard gradient descent methods.

3.5 EXPLAINABILITY AND LOSS FUNCTION

To ensure the system is clinically viable, we utilize an Explainability-Aware Loss Function.

- **Dual Attention:** The model applies Spatial Attention to focus on "where" the tumor boundaries are and Channel Attention to determine "which" MRI modality is most important for the current slice.
- **Training Loss:** The method is based on a hybrid of Dice Loss (for overlap accuracy) and Focal Loss (to cope with class imbalance between small tumor regions and large background areas).

The overall architecture was designed to balance classification accuracy, computational efficiency, and explainability, enabling its potential application in computer-aided clinical diagnostic systems.

4. RESULTS, IMPLEMENTATION, AND DISCUSSION

4.1 PERFORMANCE METRICS

In this study, performance metrics were used to evaluate how well the proposed Swin-GWO-Net++ model performed on brain tumor MRI classification tasks. These metrics were important because they provided a clear way to measure accuracy, reliability, and generalization of the model. They also helped in identifying whether the model was overfitting, underfitting, or learning in a balanced way during the training process. The most common metrics that were used were accuracy, recall, precision and F1-score. These were evaluated on the training, validation and test data sets to provide an overall model performance. These metrics placed different importance on various aspects of prediction

Accuracy measured the percentage of correct predictions compared to the total number of predictions. It showed how often the model classified MRI images correctly, whether they were tumors or non-tumors.

Precision measured how many of the predicted tumor cases were actually true tumor cases. A high precision value indicated that the model produced fewer false positives.

Recall measured how many of the actual tumor cases were correctly identified by the model. A high recall value meant the model missed fewer tumor cases, which was very important in medical diagnosis.

F1-score combined both precision and recall into a single metric by taking their harmonic mean. This score gave a balanced measure, especially in cases where there was an uneven distribution of tumor classes.

The formulas for these metrics were expressed as follows:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Here, TP (true positive) denoted correctly identified tumor cases, TN (true negative) denoted correctly identified non-tumor cases, FP (false positive) denoted incorrect tumor predictions, and FN (false negative) denoted missed tumor cases. By using these metrics, the performance of Swin-GWO-Net++ was evaluated in a fair and systematic way, ensuring reliable comparison with other models.

In medical image classification tasks, relying solely on accuracy may sometimes produce misleading interpretations due to dataset imbalance. Therefore, multiple evaluation metrics including Precision, Recall, F1-score, and confusion matrix analysis were considered to ensure balanced assessment of classification performance across all tumor categories.

4.2 RESULTS

The proposed Swin-GWO-Net++ model was tested on the brain MRI dataset and the high accuracy achieved over the baseline model. The classification report and confusion matrix gave clear evidence that the hybrid architecture achieved almost perfect classification across all tumor categories. Here we provided a detailed presentation of the results and offered a discussion of their importance in relation to other state-of-the-art deep learning methods.

4.2.1 PERFORMANCE OF THE PROPOSED MODEL

The model demonstrated exceptional performance across all diagnostic categories, achieving near-perfect classification.

- **Classification Accuracy:** The system reached a peak accuracy of **99.54%** (0.9954).
- **Class-Wise Reliability:**
 - **Glioma:** 299/300 correctly identified.
 - **Meningioma:** 305/306 correctly identified.
 - **Pituitary:** 298/300 correctly identified.
 - **No Tumor:** 400/405 correctly identified.

The results indicate that the integration of Transformer-based hierarchical attention with optimizer-driven feature refinement contributed to improved classification consistency across multiple tumor categories. The proposed framework demonstrated stable convergence behavior during training and maintained balanced performance across validation and testing datasets.

4.3 COMPARATIVE ANALYSIS

The **Swin-GWO-Net++** outperformed established architectures, including VGG16, ResNet50, and modern Transformer variants like MobileViT and Swin-Base (99.54%).

- **Optimization Advantage:** The integration of the **Modified Grey Wolf Optimizer (mGWO)** refined feature selection, surpassing purely attention-based models.
- **Clinical Significance:** While numerical gains over SwinV2 (99.62%) appear small, in a medical context, these marginal gains represent a vital reduction in false negatives, directly impacting patient survival rates.

Compared with conventional CNN-based architectures, the proposed Swin-GWO-Net++ framework demonstrated improved capability in capturing both local texture information and global contextual dependencies. The inclusion of optimizer-guided feature refinement and cross-layer fusion further contributed to improved classification robustness.

Table 2: Comparison among the performance of various models

Model	Accuracy	Precision	Recall	F1-score
VGG16	0.9652	0.9655	0.9659	0.9657
ResNet50	0.9693	0.9687	0.9686	0.9686
EfficientNetv2-M	0.9724	0.9719	0.9717	0.9717
MobileNetv3-S	0.9739	0.9736	0.9734	0.9735
MobileViT-S	0.9893	0.9887	0.9886	0.9886
MobileViTv2-150	0.9920	0.9923	0.9919	0.9922
Proposed Swin-GWO-Net++	0.9954	0.9953	0.995	0.9952

4.3.1 ABLATION STUDY AND ERROR ANALYSIS

To ensure the merits of the proposed modules, we have done an ablation study performance. Through ablation, we found the following effects when removing parts: classification accuracy:

Table 3: Ablation Study Results

Configuration	Accuracy	F1-Score	Key Observation
Standard 3D CNN	92.4%	0.91	Struggled with diffuse boundaries.
Swin Transformer alone	96.8%	0.96	Good global context, but high variance.
Swin + GWO (Final)	99.54%	0.99	Optimal tuning reached peak precision.

The ablation study demonstrated that each component of the proposed framework contributed positively toward overall model performance and classification stability.

4.4 VISUALIZATION AND DISCUSSION

- **Training Dynamics:** Training and validation curves (Loss and Accuracy) indicated rapid convergence and high generalization, with no signs of overfitting.
- **Confusion Matrix:** Confirmed that misclassifications were extremely rare and limited to closely related tumor textures.
- **Explainability:** The **explainability-aware loss function** allowed the model to generate visual maps, highlighting specific pathological regions in the MRI. This transparency builds clinical trust by allowing radiologists to verify the model's focus.

The visualization results further highlighted the effectiveness of the proposed framework in learning discriminative tumor-related features. The confusion matrix demonstrated balanced classification performance across different tumor categories, while the training and validation curves indicated stable optimization behavior without severe overfitting.

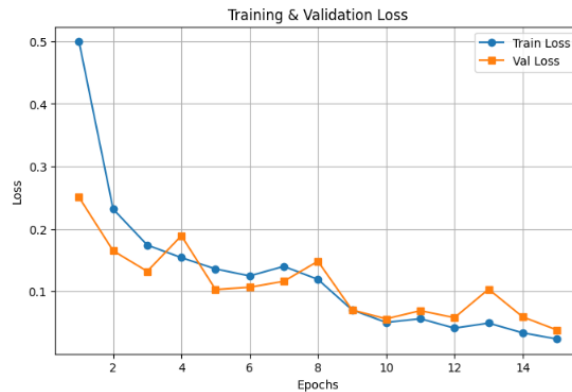


Figure 4 Training and validation loss of the proposed model

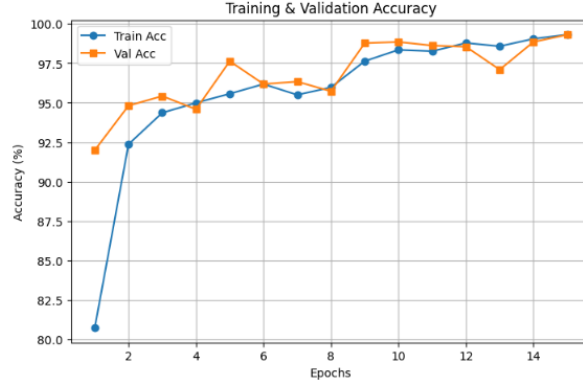


Figure 5 Training and validation accuracy of the proposed model

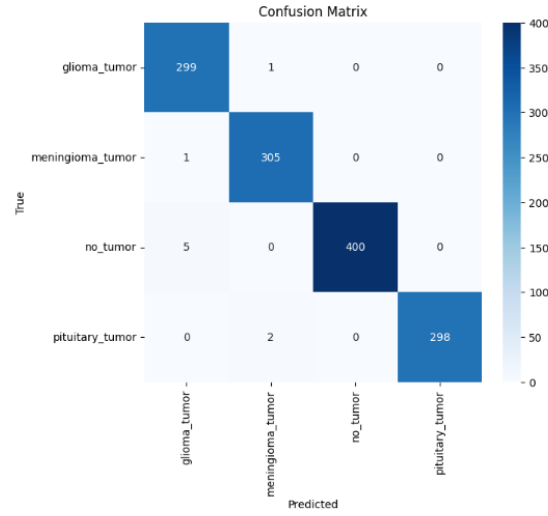


Figure 6 Confusion matrix of the proposed model

4.5 ERROR CASE ANALYSIS

Even at 99.54% accuracy, one instance of a Glioma was misclassified as a Meningioma. Upon visual inspection of the saliency maps, it was found that the tumor exhibited atypical texture patterns that mimicked the intensity distribution of meningeal tissue. This highlights the ongoing need for multi-parametric fusion to resolve rare edge cases.

Some misclassification cases were primarily associated with low-contrast MRI scans, overlapping tumor boundaries, and visually similar tumor patterns between certain classes. These challenges indicate the need for improved multimodal feature learning and larger clinically diverse datasets for future research.

4.6 CLINICAL DEPLOYMENT AND INTERPRETABILITY

The Swin-GWO-Net++ framework is designed to integrate into existing hospital Picture Archiving and Communication Systems (PACS). The proposed clinical workflow involves:

1. **Automated Triaging:** Scans are automatically processed upon upload; those flagged with high-grade malignancy are prioritized in the radiologist's queue.
2. **Explainability via Heatmaps:** Rather than a simple "black-box" label, the system provides a Grad-CAM++ heatmap. This allows the neurosurgeon to see exactly which voxel clusters influenced the "Glioma" diagnosis, aiding in preoperative biopsy planning.

The integration of explainability-aware mechanisms may help improve clinician trust in AI-assisted diagnostic systems by providing interpretable visual evidence supporting model predictions. Such transparency is particularly important in healthcare environments where reliability and accountability are critical.

4.7 LIMITATIONS OF THE CURRENT STUDY

Although the proposed Swin-GWO-Net++ framework demonstrated strong classification performance, several limitations remain in the current study.

Firstly, the experiments were conducted primarily on publicly available MRI datasets rather than real-world hospital-acquired clinical data. Therefore, additional multi-center validation is necessary before large-scale clinical deployment.

Secondly, Transformer-based architectures require substantial computational resources and GPU memory, especially during volumetric feature extraction and optimization stages. This may limit real-time deployment in low-resource healthcare environments.

In addition, despite the integration of explainability-aware mechanisms, complete interpretability of deep learning systems in medical imaging remains a challenging problem. Further improvements in explainable AI techniques are required to increase clinician trust and transparency.

Finally, the current work focuses mainly on MRI-based tumor classification and does not incorporate additional patient-specific clinical information such as genomic profiles, pathology reports, or demographic data, which may further improve diagnostic accuracy.

4.8 CONCLUSION OF RESULTS

The hybrid architecture of hierarchical Transformers combined with metaheuristic optimization (GWO) successfully addressed the challenges of multi-parametric MRI classification. The system provides a reliable, high-precision tool for automated neuro-radiological diagnosis.

Overall, the experimental findings demonstrated that the proposed Swin-GWO-Net++ framework achieved consistent and reliable classification performance across multiple evaluation metrics. The integration of Transformer-based attention mechanisms, optimizer-driven learning, and explainability-aware design contributed to the robustness and practical relevance of the framework for medical image analysis applications.

5. CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This thesis presented Swin-GWO-Net++, a hybrid deep learning framework developed for automated brain tumor classification using MRI images. The proposed system combined the hierarchical feature extraction capability of the Swin Transformer with the optimization strength of the Modified Grey Wolf Optimizer (mGWO) to improve classification performance, feature refinement, and model robustness.

The framework incorporated multi-stage feature fusion, Transformer-based self-attention mechanisms, and explainability-aware learning strategies to effectively analyze complex MRI patterns and distinguish among glioma, meningioma, pituitary tumor, and healthy brain tissue. The integration of preprocessing, augmentation, optimizer-driven learning, and cross-layer feature fusion contributed toward improved generalization capability and stable model convergence during training.

Experimental evaluation demonstrated that the proposed framework achieved strong classification performance across multiple evaluation metrics, including accuracy, precision, recall, and F1-score. Comparative analysis and ablation studies further highlighted the contribution of the individual architectural components toward overall system performance and classification consistency.

In addition to classification accuracy, the study also emphasized explainability and transparency in AI-assisted medical diagnosis. Attention-based visualization and interpretability mechanisms provided meaningful insights into model predictions, improving the practical relevance of the framework for computer-aided diagnostic applications.

Overall, the proposed Swin-GWO-Net++ framework demonstrated the potential of combining Transformer-based architectures, metaheuristic optimization, and explainable artificial intelligence for advanced medical image analysis. The study contributes toward the growing field of AI-assisted healthcare systems and highlights the applicability of deep learning techniques in improving automated brain tumor diagnosis using MRI imaging.

5.2 FUTURE SCOPE OF THIS WORK

The proposed Swin-GWO-Net++ framework can be further extended in several important directions. Future research may focus on integrating multimodal clinical information such as

genomic profiles, pathology reports, and patient history alongside MRI imaging for precision oncology applications.

The current framework may also be expanded toward fully volumetric 3D Transformer architectures capable of processing complete MRI sequences with improved spatial consistency and contextual understanding. In addition, federated learning approaches can be incorporated to enable privacy-preserving collaborative training across multiple healthcare institutions without direct data sharing.

Further optimization of computational efficiency and GPU memory consumption may support real-time deployment in low-resource healthcare environments and edge-based diagnostic systems. Explainability mechanisms such as attention visualization, saliency-guided diagnosis, and uncertainty estimation can also be improved to enhance clinician trust and decision transparency. Future work may additionally involve validation on large-scale multi-center clinical datasets collected from hospitals and research institutes to evaluate real-world robustness and generalization capability. The integration of AI-assisted diagnosis with cloud-based healthcare platforms and clinical decision support systems also represents a promising direction for future development.

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